

Reviving Intentional Facial Expressions: an Interface for ALS Patients using Brain Decoding and Image-Generative AI

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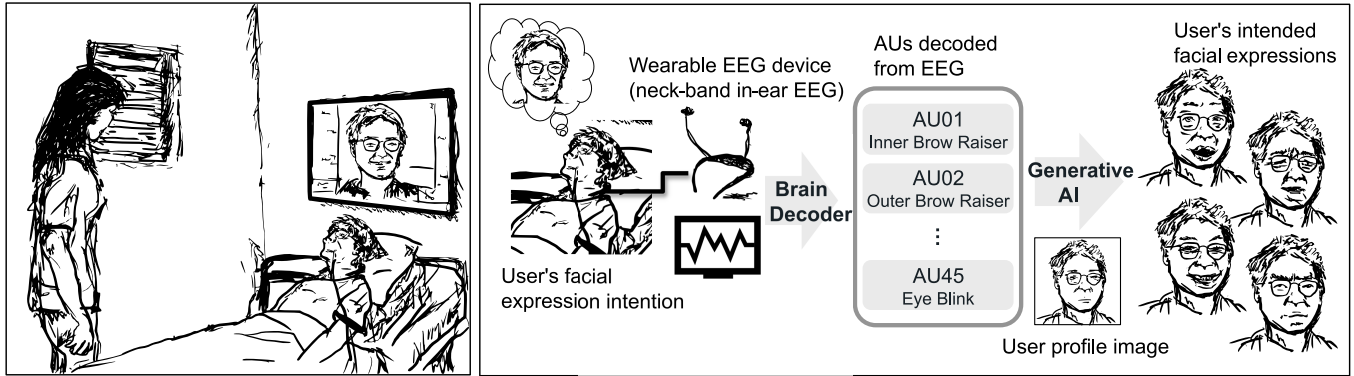


Figure 1: A concept of the proposed EEG-based facial expression generation interface. Left: A use case where a generated emotional face of an ALS patient is displayed on a bedside monitor for real-time communication. Right: A conceptual framework illustrating the proposed facial expression generation process.

Abstract

This study proposes an interface to enable non-verbal emotional communication for amyotrophic lateral sclerosis (ALS) patients with impaired face movement. By integrating EEG decoding and image-generative AI, the interface generates facial expressions based on decoding voluntary intentions rather than emotions, addressing privacy concerns. To enhance training data acquisition for decoding 17 action units (AUs), we developed a personalized approach using latent facial expression spaces and six transition animations for imitation. Experiments with participants without ALS validated the proposed method, and subsequent tests with one ALS patient showed an average AU decoding accuracy of 0.20, comparable to participants without ALS (0.23). In real-time generation tests, the ALS patient achieved a correlation of 0.40 between intended and decoded facial expressions, outperforming participants without ALS (0.30). While less accurate than invasive methods, this non-invasive, task-driven approach offers an effective solution for Augmentative and Alternative Communication in patients with severe neurological conditions.

CCS Concepts

• Human-centered computing → Accessibility; • Applied computing;

Keywords

facial expression, emotional communication, brain computer interface (BCI), electroencephalogram (EEG), brain decoding, amyotrophic lateral sclerosis (ALS)

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1 Introduction

Brain-computer interface (BCI) technology facilitates communication between the brain and computers by recording neural activity [18]. In the fields of medicine and life sciences, it shows significant potential as an augmentative and alternative communication (AAC) system to enhance nonverbal communication and improve the quality of life (QoL) [6, 8, 12, 26]. AAC systems are particularly crucial for individuals with locked-in syndrome (LIS), such as those with amyotrophic lateral sclerosis (ALS), who retain full cognitive function yet lose nearly all means of communication due to severe motor impairments [20, 27]. Moreover, for patients with completely LIS, who lack even minimal motor abilities, BCI-based AAC technology serves as the final viable means of communication [19, 24].

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Although most AAC systems prioritize verbal communication, several studies on AAC systems have shown the importance of nonverbal communication in conveying emotions [6, 12, 26]. However, existing systems require patients to consciously express their emotions by selecting their emotional states through gaze input systems or BCIs. There are several drawbacks to this approach. First, using these systems is labor-intensive, and the additional task of selecting an emotional state along with composing a verbal message burdens the user. Second, the input process is time-consuming, making it difficult to respond promptly during interactions with their surroundings, such as returning a smile in response to a call. Lastly, emotions are inherently complex and challenging to interpret precisely, as discrete categorical or linguistic representations often fail to fully capture the nuances of actual emotional states.

The use of EEG in emotion-based systems also raises significant privacy concerns, as EEG data reflects involuntary brain activity that could unintentionally reveal private information. Recent legal frameworks, including the General Data Protection Regulation (GDPR) (Regulation (EU) 2016/679) and the EU's Artificial Intelligence Act (AI Act), emphasize the importance of protecting neural data while supporting innovation in neurotechnology [9, 22]. These regulations aim to ensure data privacy while enabling advancements in communication technologies for vulnerable populations.

In this context, we propose a novel, privacy-conscious method for generating facial expressions from EEG signals, focusing on decoding users' volitional facial expressions instead of emotional states. This approach prevents the unintended disclosure of private emotions, offering users greater control over their communication. A survey of ALS patients highlighted the demand for interfaces that enable volitional expression. Building on this, our method generates facial expressions through EEG decoding, allowing individuals with severe motor impairments to achieve meaningful and ethical nonverbal communication.

2 Related Works

2.1 EEG-based Emotion Recognition

EEG-based emotion recognition has recently gained attention, with two main approaches: traditional classifiers, which rely on statistically derived EEG features, and deep learning-based end-to-end methods, which integrate feature extraction and classification. Traditional methods compute features such as standard deviation, root mean square, and power spectral density [11, 15], as well as advanced features like wavelet transformations [10], energy [3], and fractal dimensions [16]. These features enhance data representation and classifier performance. Deep learning methods, such as EEGNet, utilize CNNs for accurate classification across BCI paradigms [13], while EEG2Vec employs VAEs to encode EEG signals into 1,000 latent variables for affective state estimation and EEG synthesis [2].

Although over 80% accuracy has been reported for simple emotions like valence and arousal [14], ethical concerns arise with systems that recognize emotions without user consent. This paper addresses these concerns by proposing an AAC system focused on volitional emotional expression rather than automatic emotion recognition.

2.2 EEG-based Facial Expression Estimation

Emotions are communicated non-verbally through facial expressions and gestures [5]. Recognizing facial expressions from EEG may enable detecting the intended emotional expressions of the user.

Salari et al. proposed a BCI system designed to classify facial expressions for the intended display of emotions, demonstrating its potential for emotional communication in individuals with motor impairments [23]. The reported accuracy is relatively high at 72%. However, it uses electrocorticography (ECoG), requiring implanted electrodes, limiting its practical application. Additionally, it classifies only five basic emotions, leaving its ability to detect subtle expressions uncertain.

Shimizu et al. proposed a BCI system using Action Unit (AU)-based facial expression recognition with consumer-grade EEG [25]. AUs are facial muscle movements categorized in the Facial Action Coding System (FACS) [7]. The system achieved over 60% accuracy for recognizing fifteen emotional states and expression intentions using eight AUs. However, it simultaneously detects emotional states and facial expressions from EEG during emotion induction, limiting its ability to estimate expressions the user intends to convey without emotions or with different emotional states. This paper investigates AU-based facial expression recognition from EEG signals without emotion estimation and explores an efficient way to acquire EEG signals for training a brain decoder.

3 Proposed EEG-based Intentional Facial Expression Generation Interface

Figure 1 shows the overall framework of the proposed facial expression generation system using EEG-based intentional facial expression decoder and facial expression generation AI technology. The proposed method uses AUs rather than emotion categories to represent facial expressions and supports arbitrary detailed facial expressions. Therefore, the proposed method decodes the intention of facial expressions in AU units in real-time from EEG signals and generates the user's facial animations through AUs.

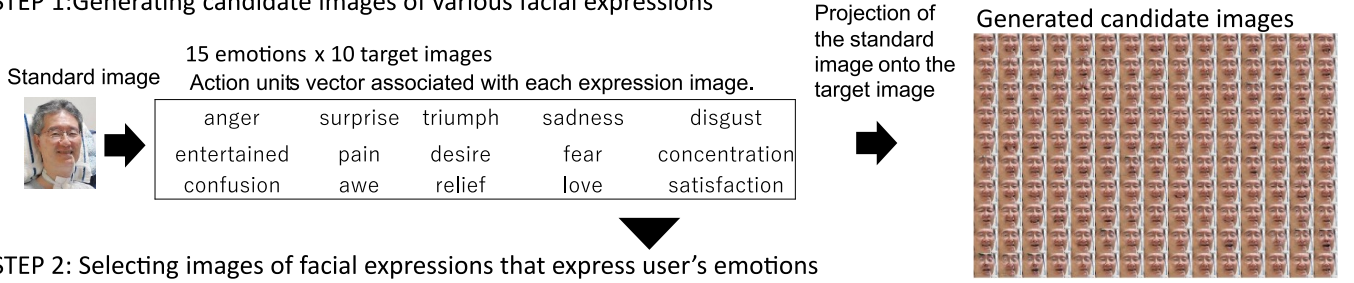
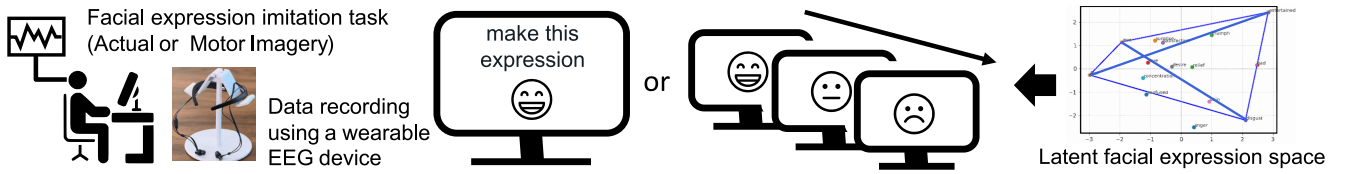
To enable AU-based facial expression synthesis, we employed GANimation [21], a GAN-based framework designed to generate anatomically-aware and realistic facial animations by manipulating expressions through AUs. The GANimation model was trained over 70 epochs on the CelebA dataset [17]. Before training, the images in the dataset were aligned, and AU vectors were extracted using OpenFace [1]. The trained GANimation model utilizes 17 AU annotations¹.

The proposed EEG-based AU decoders, namely EEG2AU models, estimate the intention of facial expression based on the last 2 seconds of EEG signals. These decoders shall be pre-trained for each user, and the method of acquiring the training data and the decoding algorithm are described in detail in the following subsections.

3.1 Training data acquisition

To generate personalized facial expressions reflecting emotions and corresponding to individual EEG, training data is collected in three steps (Figure 2). First, a basic user profile image is created

¹AU01, AU02, AU04, AU05, AU06, AU07, AU09, AU10, AU12, AU14, AU15, AU17, AU20, AU23, AU25, AU26, AU45

STEP 1: Generating candidate images of various facial expressions**STEP 2: Selecting images of facial expressions that express user's emotions****STEP 3: Training (Recording) session****Figure 2: Data collection procedure to train EEG-based AU decoders.**

(via webcam or upload) and aligned using OpenFace [1]. Then, using this image and the pre-trained GANimation model, 150 facial expression images representing 15 emotions (10 variations per emotion) are generated. Based on Cowen and Keltner's study [5], 15 common emotions are selected², and 10 representative images per emotion are chosen as targets in GANimation to create diverse facial expressions.

In the second step, users selected an image from the candidate facial expressions to represent their own emotion. The interface allowed participants to choose the image that best matched their facial expression for each of the 15 emotions.

The third step involved acquiring EEG data to train the EEG2AU model. To reduce the data acquisition burden, two methods were compared. The study also included two experimental conditions: one where participants physically made facial expressions (Actual) and another where they only imagined them (Motor Imagery).

Method 1: With static image for each emotion.

The 15 selected facial expression images were presented for 12 seconds each. Users were instructed to either imagine or create the facial expression corresponding to each image. The session was repeated three times, with each trial consisting of 15 emotions. The imagination and actual facial expression conditions were alternated, and the emotion order was randomized for each participant.

Method 2: With video that moves between facial expression spaces.

Principal Component Analysis (PCA) reduced the 17-dimensional AU vectors of the selected facial expression images for 15 emotions to a 2D space. Four points representing specific

emotions were chosen to maximize the enclosed area using the Shoelace Formula, defining six emotion pairs along the four sides and two diagonals. This focused the learning process on transitions between these pairs, reducing the training trials from 15 emotions to six pairs. GANimation generated smooth transitions between these expressions, creating 100-frame animations for each pair.

The videos were shown to users, who were instructed to imagine or imitate the transitions. Each trial consisted of a 2-second video of the transition followed by a 2-second still image of the target expression. Each session included two repetitions of five round-trips for each emotion pair, alternating between imagination and actual expression. The condition order was counterbalanced, and emotion sequences were randomized for each participant. This optimization reduced training data acquisition time by 11%, improving efficiency in covering the facial expression space.

3.2 EEG-based AU decoder model training

During the training session, users' brain activities were recorded during the facial expression imitation task. To ensure general usability, a wearable EEG device (VIE ZONE, VIE Corporation, in-ear EEG with reference electrodes on the neck) was used instead of a burdensome cap-type device. This device recorded brain activity from the left and right ear canals at a 600 Hz sampling rate.

The EEG data was segmented into 2-second windows, with PSD calculated in the 4–40 Hz range at 0.5 Hz increments, sliding windows at 1-second intervals. Despite the risk of data leakage, this method was chosen to ensure sufficient data for training. Noise was quantified for each segment using a preset algorithm, and segments exceeding three times the standard deviation were excluded.

²anger, surprise, triumph, sadness, disgust, entertained, pain, desire, fear, concentration, confusion, awe, relief, love, and satisfaction

[4]. This threshold, more lenient than [4], allowed for the inclusion of electromyography (EMG) signals generated during facial expressions, applicable to both ALS and non-ALS participants. The Each frequency's power was standardized per subject. EEG features included the power of each frequency for the left ear, right ear, and their difference, extracted per window. PCA compressed these features to 100 dimensions for further processing.

Each segmented EEG dataset was paired with 17-dimensional annotations representing the intended intensity of each AU during recording. We used the intended AUs from the presented facial image rather than actual facial movements, even in the condition where the user physically attempted to make facial expressions. This approach focuses on decoding the intention to produce specific expressions, as patients with LIS—the study's target group—cannot fully execute facial expressions.

A separate EEG2AU model was trained for each AU to predict its intensity using compressed EEG features. A generalized linear regression model with the least absolute shrinkage and selection operator (LASSO) was adopted for real-time decoding, with the parameter λ optimized via 10-fold cross-validation.

4 Experiments

We conducted experiments to evaluate the feasibility of the proposed interface, focusing on the decoding accuracy of the EEG2AU models and comparing different methods for acquiring training data. As preparatory experiments, eight participants without ALS participated in Method 1, and nine in Method 2. Finally, one ALS patient participated in the validation phase.

This interface is designed for users with ALS and other conditions that impair facial muscle movement. These users may struggle with actual muscle movements during the learning process, making imagined movements a viable option for training data. However, brain activity may be more interpretable when users attempt to create facial expressions using any remaining muscle function, even if the movement is incomplete. To assess this, preparatory experiments with participants without ALS compared decoding accuracy under two conditions: actual facial expression attempts (Actual) and imagined imitation of the presented expressions (Motor Imagery). The approach with higher accuracy was then tested with participants diagnosed with ALS.

To evaluate the decoding accuracy of the EEG2AU model, 80% of the training data was used for training and 20% for testing. This evaluation assessed whether the model could accurately decode the intention to generate facial expressions, represented as AU units, from EEG signals. We evaluated accuracy by calculating the correlation coefficient for each AU between the decoded intensities and those of the target emotional images. For AUs where effective training was not feasible, only the intercept was retained, all regression coefficients (β) were set to 0, and the accuracy for those AUs was recorded as 0.

Additionally, a separate test session was conducted using EEG2AU models trained on the full dataset to assess the interface's performance in generating facial expressions. During this session, 15 emotional images were randomly presented, each for 24 seconds.

We evaluated accuracy by calculating the correlation coefficient between the 17-dimensional AU intensity vector decoded in real-time from EEG and that of the target emotional image.

4.1 Results with participants without ALS

Figure 3 presents a comparison of decoding accuracy per AU between the Motor Imagery and Actual conditions, based on the correlation coefficient between decoded and target AU intensities. Although absolute decoding accuracy and differences between conditions varied across AUs, the Actual condition outperformed Motor Imagery for many AUs and demonstrated significantly higher accuracy on average (Motor Imagery: 0.07; Actual: 0.24, $p < .05$). These results support the assumption that the Actual condition, where subjects physically attempt facial expressions, generates a stronger intention to produce expressions. Additionally, EMG signals from facial muscles likely mixed into the EEG data, making the recorded signals more interpretable.

Next, to compare the efficiency of training data acquisition for decoding the intention to generate facial expressions from brain waves, we evaluated Method 1 and Method 2 under the Actual condition. Using the non-parametric Wilcoxon signed-rank test, we found that despite Method 2 requiring 11% less training data, its accuracy was not inferior to Method 1. The average accuracy was 0.14 for Method 1 and 0.24 for Method 2.

As shown in Figure 4, Method 2 demonstrated higher correlation for certain AUs. For example, for AU20 (pulling the lips to the side), the accuracy was 0.08 for Method 1 and 0.33 for Method 2 ($P < .05$). These results suggest that the video-based facial expression imitation task in Method 2 was easier for users and enabled more efficient data acquisition by effectively covering the facial expression space for each individual.

4.2 Validation on One ALS Patient

The optimal training process identified in the preparatory experiments with non-ALS participants, i.e., the video-based task of actually creating facial expressions (Actual condition using Method 2), was used to validate the proposed system with an ALS patient. The participant was a 59-year-old male ALS patient who was diagnosed with ALS 13 years ago. Although ALS patients have difficulty expressing facial expressions, their facial muscles are not entirely immobile. The participant was instructed to imitate facial expression changes presented as animations, following the same procedure as in the non-ALS experiments. Each trial consisted of 4 seconds $\times$ 5 repetitions $\times$ 6 emotion pairs, repeated twice, for a total duration of 8 minutes.

Table 1 shows the AU-based accuracy of the EEG2AU models on test data from the training session, evaluated using correlation coefficient between decoded and target AU intensities. The ALS patient achieved an average decoding accuracy, evaluated using the correlation coefficient, of 0.20, which is comparable to non-ALS participants (0.23). This result indicates that AU-based motor intentions can be decoded from EEG even for ALS patients with limited facial expression ability. Training failed for AU01, AU02, AU17, AU26, and AU45, aligning with the participant's post-training comment: "Frowning was easy, but eye-widening was difficult." These results suggest that model training depends not only on

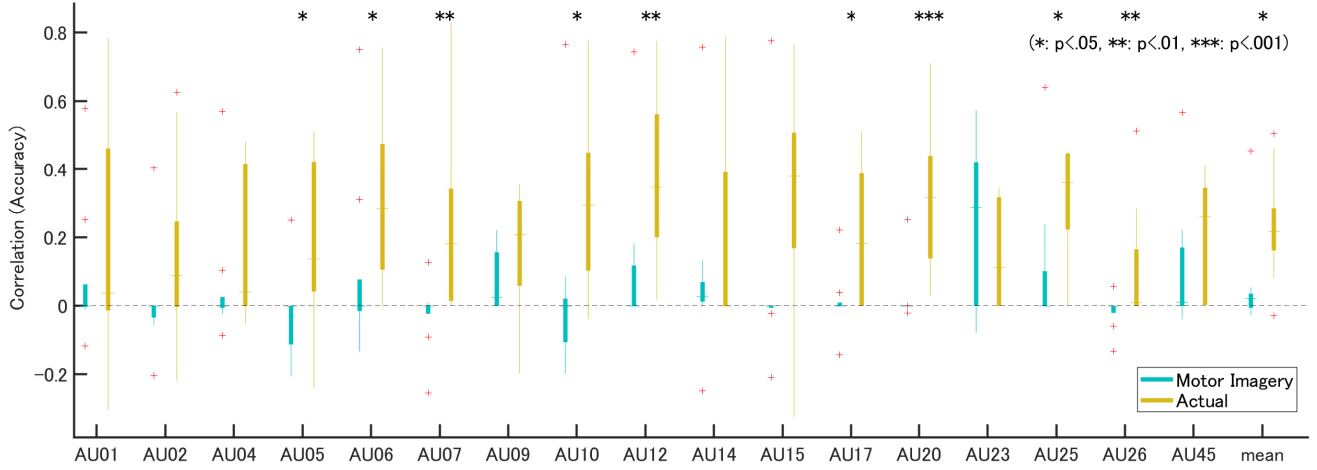


Figure 3: Comparison of Motor Imagery conditions and Actual conditions using AU intensity correlation.

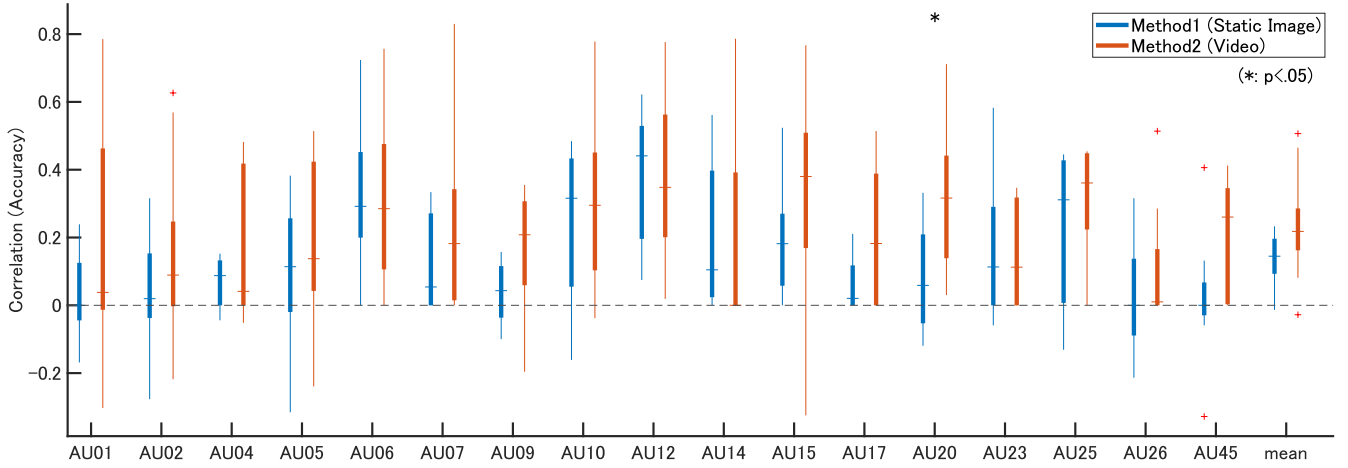


Figure 4: Comparison of Method 1 (still images) and Method 2 (videos) using AU intensity correlation under actual conditions.

physical facial expression ability but also on the user's capacity to exhibit motor intentions in AU units.

Table 2 presents the emotion-based accuracy for each participant during the test sessions, evaluated using correlation coefficient between decoded and target AU intensity vectors. The average correlation of the ALS participant was 0.40, which was comparable to or exceeded the average correlation of the non-ALS participants (0.30). Although further improvements in accuracy are needed, the proposed interface demonstrated the ability to generate facial expressions with varying accuracy depending on the emotion. These results confirm its potential to generate intended facial expressions using a consumer-grade EEG device. In the post-test interview, the ALS participant reported difficulty expressing fear, and indeed, the accuracy for fear was low (0.13). This suggests a correlation between the difficulty of creating subjective facial expressions and the accuracy of decoding facial expression intentions.

5 Conclusion

This research aimed to enable ALS patients to communicate non-verbally through emotional facial expression generation technology utilizing EEG and image-generative AI. We designed and implemented an interface that uses a simple, wearable EEG device to decode the user's intended facial expressions and generate corresponding facial expression images in real-time. Compared to existing interfaces, a key innovation of our approach lies in decoding the user's voluntary facial expression intentions—such as "I want to make this kind of facial expression"—rather than attempting to infer emotional states directly from brain activity. This approach addresses privacy concerns associated with unintended emotion decoding and is anticipated to serve as a valuable interface for individuals with neurological conditions, such as ALS, who experience difficulty in freely moving their facial muscles. By focusing on decoding AUs from brain activity, we also optimized training data collection through a personalized latent facial expression space,

Table 1: AU-based correlation coefficients on the test data in the training session (Actual conditions with Method 2).

Participant	Action Unit Number																	ave.
	01	02	04	05	06	07	09	10	12	14	15	17	20	23	25	26	45	
non-ALS1	0.11	0.09	0.04	0.22	0.28	0.00	0.08	0.20	0.27	0.15	0.29	0.18	0.36	0.11	0.36	0.12	0.36	0.19
non-ALS2	-0.30	-0.22	-0.05	-0.24	0.29	0.02	-0.20	-0.04	0.21	0.00	-0.19	0.02	0.03	0.07	0.00	0.01	0.12	-0.03
non-ALS3	0.03	0.14	0.00	0.10	0.35	0.14	0.22	0.30	0.35	0.00	0.30	0.34	0.12	0.31	0.34	0.51	0.30	0.23
non-ALS4	0.73	0.63	0.37	0.50	0.76	0.32	0.35	0.78	0.78	0.79	0.77	0.00	0.71	0.34	0.45	0.01	0.34	0.51
non-ALS5	0.04	0.14	0.00	0.14	0.14	0.20	0.21	0.14	0.18	0.00	-0.32	0.00	0.27	0.00	0.00	0.00	0.26	0.08
non-ALS6	0.79	0.57	0.48	0.51	0.63	0.83	0.29	0.45	0.61	0.41	0.51	0.51	0.32	0.00	0.30	0.29	0.41	0.47
non-ALS7	0.00	0.00	0.40	0.00	0.00	0.41	0.35	0.00	0.02	0.00	0.38	0.39	0.35	0.35	0.45	0.13	0.00	0.19
non-ALS8	-0.05	-0.01	0.48	0.06	0.42	0.18	0.17	0.46	0.46	0.39	0.45	0.00	0.14	0.24	0.45	0.00	0.00	0.23
non-ALS9	0.37	0.00	0.00	0.40	0.00	0.00	0.00	0.38	0.55	0.00	0.51	0.40	0.70	0.00	0.40	0.00	0.00	0.22
non-ALS mean	0.19	0.15	0.19	0.19	0.32	0.23	0.17	0.30	0.38	0.19	0.30	0.20	0.33	0.16	0.31	0.12	0.20	0.23
ALS	0.00	0.00	0.15	0.30	0.28	0.39	0.32	0.27	0.28	0.26	0.30	0.00	0.37	0.33	0.13	0.00	0.00	0.20

Table 2: Emotion-based accuracy in the test session (Actual condition with Method 2), evaluated using correlation coefficient between decoded and target AU intensity vector .

Participant	Emotion															
	anger	surprise	triumph	sad	disgust	ent'd*	pain	desire	fear	conc. [†]	confused	awe	relief	love	sat. [‡]	ave.
non-ALS1	0.45	0.24	0.37	0.26	0.46	0.40	0.09	0.54	0.20	-0.22	0.67	0.25	0.58	0.67	0.46	0.36
non-ALS2	0.13	0.38	0.31	-0.08	0.26	0.27	0.35	0.31	0.69	0.30	0.45	0.05	0.01	-0.08	0.02	0.23
non-ALS3	-0.07	0.78	0.00	0.80	0.02	0.38	0.36	-0.11	0.88	-0.41	-0.14	0.60	0.45	0.28	-0.08	0.25
non-ALS4	-0.03	0.61	0.31	0.68	0.30	0.88	0.40	0.53	0.10	0.53	0.37	0.18	0.41	0.81	0.67	0.45
non-ALS5	0.34	0.22	0.27	-0.16	0.59	0.46	0.65	0.01	0.12	-0.36	0.32	0.19	0.11	0.26	-0.01	0.20
non-ALS6	0.07	0.27	-0.22	0.06	-0.14	0.79	0.37	0.52	0.27	0.53	0.43	0.67	-0.20	0.18	-0.16	0.23
non-ALS7	0.46	0.09	0.60	-0.25	0.43	0.65	0.53	0.53	-0.06	0.29	0.62	0.06	-0.16	0.73	0.70	0.35
non-ALS8	0.38	0.76	0.44	0.20	0.41	0.16	0.26	0.18	0.52	0.53	0.43	0.28	-0.47	0.25	0.56	0.33
non-ALS9	0.52	0.58	0.57	0.05	0.02	0.00	0.18	0.58	0.31	0.12	0.54	0.55	0.04	0.53	0.57	0.34
non-ALS mean	0.26	0.42	0.34	0.21	0.28	0.51	0.34	0.33	0.32	0.14	0.39	0.34	0.12	0.43	0.33	0.30
ALS	0.33	0.30	0.69	0.51	0.44	0.63	0.24	0.17	0.13	0.09	0.22	0.59	0.41	0.70	0.52	0.40

*entertained, [†]concentration, [‡]satisfaction

where users imitate six facial expression transitions, reducing the burden on ALS patients.

The ALS patient who participated in our study expressed strong interest in the interface and its potential, noting that even a basic system for selecting predefined facial images would greatly enhance communication. He also expressed interest in an automatic facial expression generation feature based on brain signals. Additionally, he suggested focusing on a small set of common expressions (e.g., smiling, surprise, confusion, anger) with improved accuracy and emphasized the importance of enabling variations within specific expressions, such as different types of smiles, to better convey nuanced emotions.

While consumer-grade EEG devices have lower signal resolution and accuracy than invasive brain measurement systems, our approach demonstrates that thoughtful task design can mitigate these limitations. The ALS patient emphasized the importance of non-invasive and user-friendly devices, even if their accuracy is slightly lower than surgically implanted systems. We anticipate that our approach will contribute significantly to advancements in AAC technologies for individuals with ALS and similar conditions.

6 Ethical Impact Statement

The experiments conducted in this study were reviewed and approved by the local ethics committee (approval number: 2023-1, data of approval: 14th July 2023). All participants, including one

ALS patient, were informed of the experimental protocols and consented before the experiment. Additionally, the experiment on the ALS patient was conducted in the presence of their family and a care manager. He also agreed to disclose his images in this paper publicly.

The proposed technology aims to enhance the communication and quality of life of ALS patients by decoding their intended facial expressions rather than predicting emotional states, thereby addressing privacy concerns related to unintended emotion decoding. However, since brain activity is difficult to control deliberately, there is a risk of users unintentionally generating incorrect expressions. To ensure practical usability, it is crucial to improve decoding accuracy and incorporate mechanisms that allow explicit user control.

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A Selected Facial Images

Figure 5 shows a list of the facial images that the participants in the experiment selected as the most appropriate facial expression to represent their emotions at the 2nd step of the training data acquisition described in Section 3.1. It can be seen that different facial expressions are selected for each participant, even when they express the same emotion.

It can also be confirmed that GANimation can generate high-quality and naturalistic facial expression images while preserving the person's identity. Since the proposed interface supports any combination of AUs of any intensity decoded from EEG, facial expression images other than those illustrated here are also generated but of approximately the same quality in real-time.

B Latent Facial Expression Spaces

Figure 6 shows the latent facial expression spaces, which are calculated in Method 2 of the 3rd step of the training data acquisition process described in Section 3.1. The latent space was personalized based on the participant's selected facial image.

The blue rectangle is the area obtained using the Shoelace Formula, and instructional videos for training data acquisition were created along each side and diagonal of this rectangle. Although some emotions cannot be covered, it can be confirmed that the areas corresponding to most emotions are generally covered.

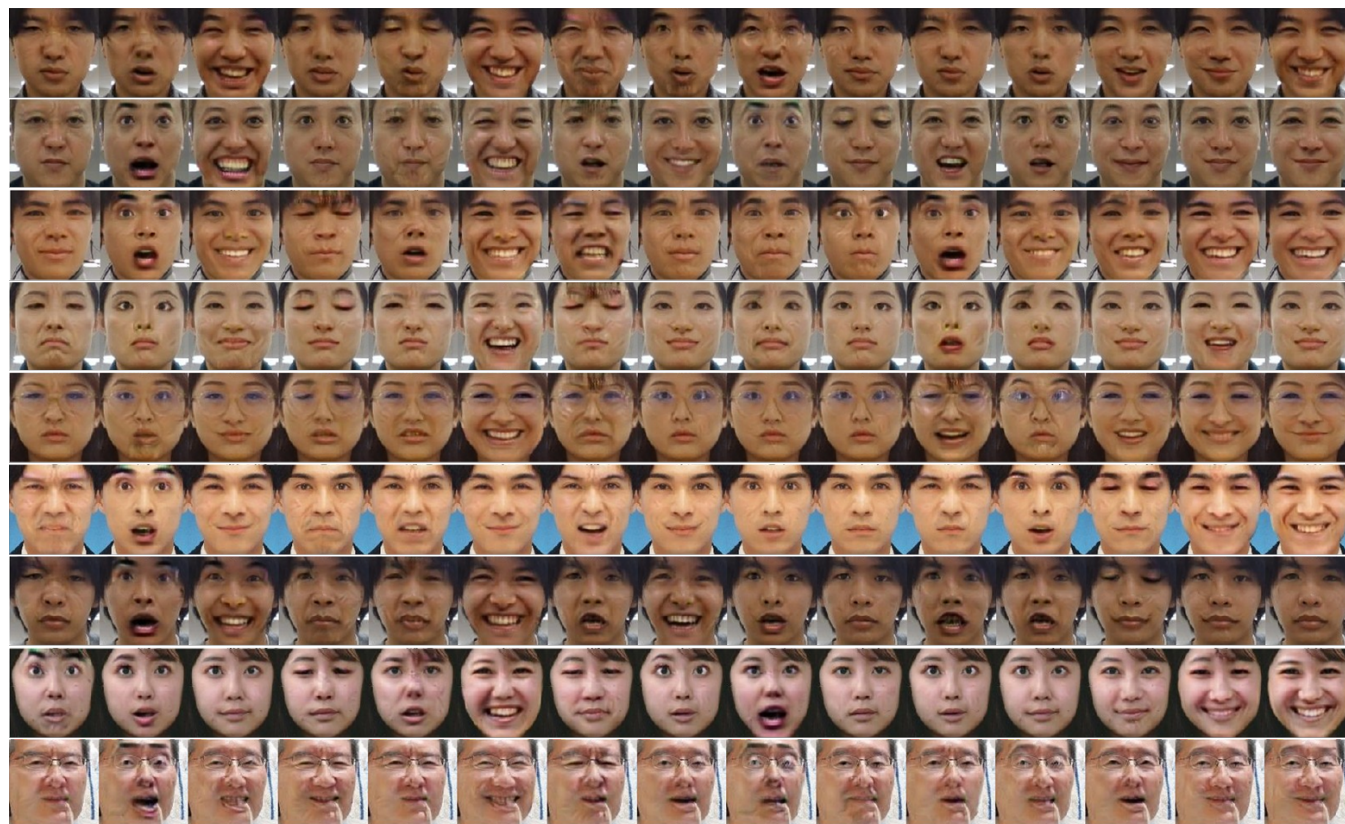


Figure 5: Emotional facial expression selected by participants. Each row corresponds to one participant in the experiment, and from left to right, each column represents anger, surprise, triumph, sadness, disgust, entertained, pain, desire, fear, concentration, confusion, awe, relief, love, and satisfaction. Note that only images of participants who gave permission for publication are shown.

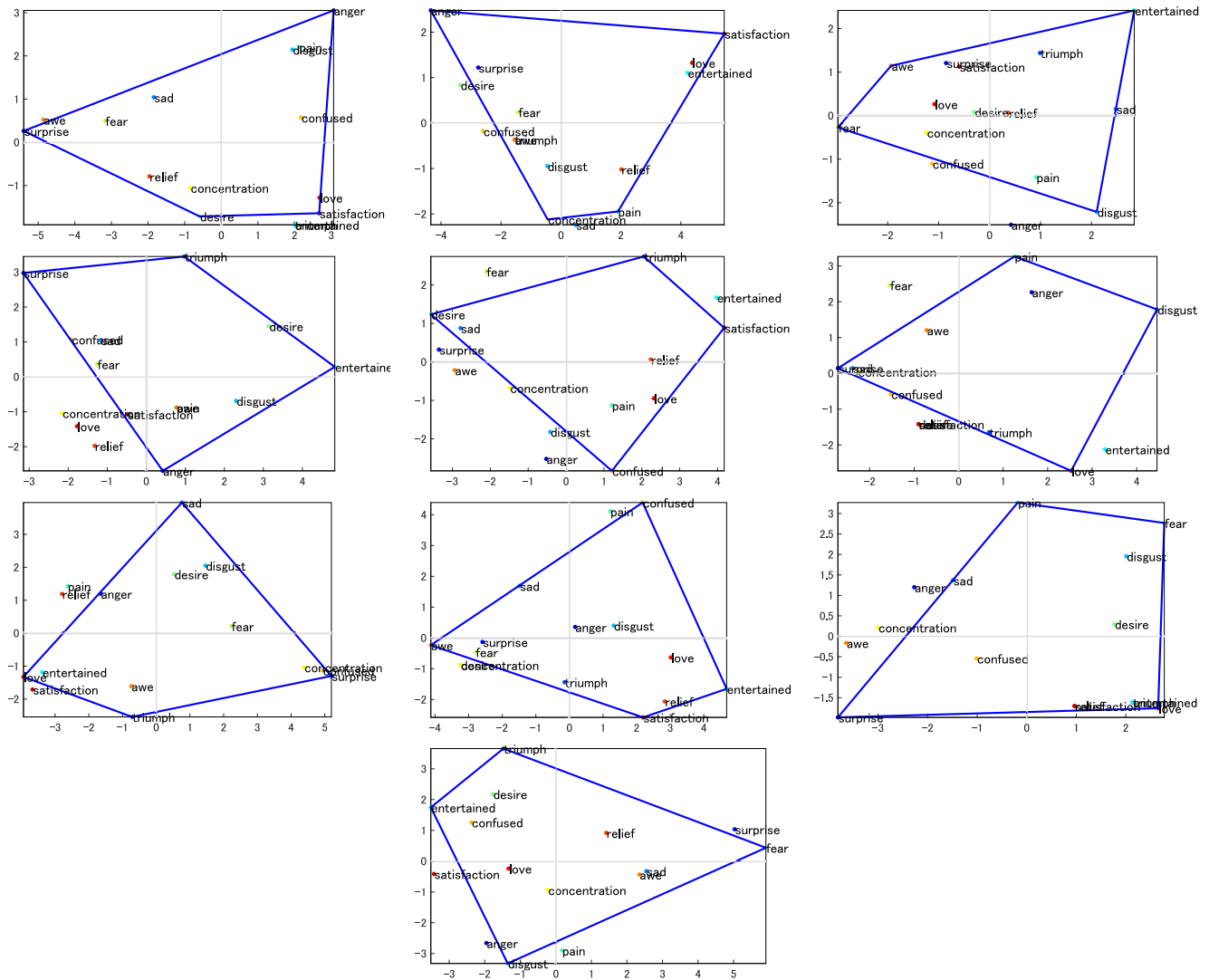


Figure 6: Latent Facial Expression Space personalized for each participant. The horizontal axis corresponds to the first principal component and the vertical axis to the second principal component.